Unit 2.1 Printables: Teacher Answer Key

One printable per page, in the same order as the student printables. Answers are in the bordered KEY box; where students write in a table, the table is shown filled in (answers in bold italic).

Siamese cat data

Circle the 4 dark body parts. Write W next to the warm parts and C next to the cool parts.

Data 1: Temperature and fur color

Body part of an adult Siamese cat	Skin temperature (°C)	Fur color
Back (trunk)	36	Cream
Belly	36	Cream
Upper leg	34	Cream
Ear	28	Dark brown
Nose and face mask	30	Dark brown
Paw	28	Dark brown
Tail tip	27	Dark brown

Simplified class data set. Real temperatures vary.

Data 2: Litters of kittens

Litter	Mother	Father	Dark-pointed (Siamese pattern) kittens	Solid-colored kittens
1	Siamese pattern	Siamese pattern	5	0
2	Siamese pattern	Siamese pattern	4	0
3	Solid	Solid	1	4
4	Solid	Solid	2	3

Simplified class data set modeled on real litter patterns.

KEY Circle: ear (28 °C), nose and face mask (30), paw (28), tail tip (27): all dark brown, all marked **C** (cool). Back (36), belly (36), upper leg (34): cream, marked **W** (warm).
Litters: two Siamese-pattern parents had only Siamese-pattern kittens (5 of 5 and 4 of 4). Two solid parents can still have Siamese kittens (litters 3 and 4: 1 and 2), so a kitten gets genes from both parents.

Reading: DNA, genes, and chromosomes

Annotate: underline each key word the first time you see it (DNA, gene, protein, chromosome). Draw an arrow from DNA to gene to protein. Put a ? next to one thing you still wonder.

Every cell in a cat stores instructions in **DNA**. A **gene** is a section of DNA with the instructions for one **protein**. Proteins do the jobs of the cell, like making the dark pigment in fur.

DNA is packed into **chromosomes**. A chromosome is one long DNA molecule packed with proteins, and it holds many genes. A kitten gets one set of chromosomes, and so one set of genes, from each parent. That is why two solid-colored parents can still have a Siamese kitten: each parent can pass on a changed copy of the gene.

Big question: How can a difference in DNA change which parts of a cat are dark?

KEY Underline: DNA, gene, protein, chromosome (first use of each). **Arrow:** DNA → gene → protein (→ job, such as pigment). **?:** any honest question; good ones ask how two solid parents can have a Siamese kitten, or what the changed gene does.

Reading: What is DNA made of?

Annotate: draw the shape next to each part (pentagon = sugar, circle = phosphate, rectangle = base). Box the base-pairing rule.

DNA is a **double helix**, shaped like a twisted ladder. It is built from **nucleotides**. Each nucleotide has three parts: a **sugar**, a **phosphate**, and a **base**.

The sugars and phosphates make the sides of the ladder (the backbone). The bases make the rungs. There are four bases: A, T, C, and G. **A pairs with T. C pairs with G.**

The order of the bases is the message. A cat's DNA has billions of bases, and a gene is one stretch of them.

Solid cat: A T G G G A C T G T T T
Siamese cat: A T G A G A C T G T T T *(made-up letters that show the idea)*

KEY Shapes: pentagon next to sugar, circle next to phosphate, rectangle next to base. **Box:** "A pairs with T. C pairs with G." The two cat sequences differ at one letter: the 4th (G in the solid cat, A in the Siamese cat).

Base key and code table

Keep this card on your lab tray. Read the copy three letters at a time.

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Pairing rule: A with T, C with G.

KEY Reference card; nothing to fill in. Lab answers are on the lab sheet key page.

Card sort: what can one letter do?

For each card, write 1, 2, or 3 and a reason. Card B is done for you. 1 = changes an amino acid · 2 = changes whether the protein is made · 3 = no noticeable change

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Reason bank: No gene is involved. · The protein is not made. · The amino acid stays the same. · The amino acid changes from ___ to ___.

Card	What changed	1, 2, or 3	Reason
A	Siamese cat: GGA becomes AGA in the pigment gene.	***1***	***The amino acid changes from Gly to Arg.***
B	GAG becomes GAA.	3	The amino acid stays the same.
C	CTG becomes CCG.	***1***	***The amino acid changes from Leu to Pro.***
D	A change in DNA with no gene and no switch.	***3***	***No gene is involved.***
E	A change in a gene's switch, so the gene is never turned on.	***2***	***The protein is not made.***
F	Human gene: GAG becomes GTG.	***1***	***The amino acid changes from Glu to Val (as in sickle cell).***

KEY Card B is the worked example. Category 1 = A, C, F; category 2 = E; category 3 = B, D.

Same DNA, different jobs

Highlight what is the same in cells P and Q. Then fill in the temperature table. We do the ear together.

Two kinds of cells

Cell type	Genes in the cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Where does the changed protein work? (cooler than 33 °C)

Body part	Skin temperature (°C)	Cooler than 33 °C? Protein works?	Fur color
Ear	28	***yes***	***dark***
Nose	30	***yes***	***dark***
Paw	28	***yes***	***dark***
Tail tip	27	***yes***	***dark***
Upper leg	34	***no***	***cream***
Back	36	***no***	***cream***
Belly	36	***no***	***cream***

KEY Highlight: both cells have genes 1, 2, 3, 4 (the same DNA). They make different proteins (1 and 2 vs. 3 and 4), so they have different jobs.

Case data

In each case, circle the DNA change, underline the protein change, and box the effect on the body.

Case 1: Siamese and solid cats

	Solid cat	**Siamese cat**
Gene for the pigment-making protein	Usual sequence	One letter different
Amino acid chain	Usual chain	One amino acid different
Where the protein works	At every body temperature	Only in skin cooler than about 33 °C
Fur color	Dark all over	Dark ears, nose, paws, tail; cream body
Temperature of ear / back	28 / 36 °C	28 / 36 °C

Case 2: A human gene

Person	**Gene**	**Protein**	**Symptoms**
1	Normal	Works	None
2	One base changed	One amino acid different; works poorly	Yes

KEY Case 1: circle “One letter different”; underline “One amino acid different”; box “Dark ears, nose, paws, tail; cream body” (protein works only in skin cooler than about 33 °C). **Case 2:** circle “One base changed”; underline “One amino acid different; works poorly”; box “Yes” (symptoms).

Write It: two Siamese kittens

Circle the two numbers you will use as evidence. Then check off each part of your model as you add it.

Two Siamese kittens are raised from birth, one in a warm room and one in a cool room. At 12 weeks:

	Kitten raised warm	**Kitten raised cool**
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Model checklist

☐ DNA with the gene ☐ the one-letter change ☐ the amino acid chain with one amino acid different ☐ the protein working only in cool skin ☐ the dark points

KEY Circle: ear 26 °C, dark brown (cool room) and ear 34 °C, light tan (warm room). Backs 36 and 35 °C are cream in both. **Model checklist:** all 5 boxes should be checked on the revised model.

Gummy DNA Lab: Build, Copy, and Read a Gene

KEY

- **Part A** colors in order: yellow, red, green, green, green, yellow, orange, red, green, red, red, red.
- **Part B** partner strand: A T G G G A C T G T T T. The partner strand carries the coding letters; it matches the copy the cell makes from the template.
- **Part C** copy: A T G G G A C T G T T T. Chain: **Met – Gly – Leu – Phe**. The gene's message is copied from the template strand, and the copy is read three letters at a time to link amino acids in order.
- **Part D**: the changed template TAC TCT GAC AAA gives the copy ATG AGA CTG TTT: **Met – Arg – Leu – Phe**. One amino acid changed (the second, Gly to Arg). Yes, a different chain can make a protein that works differently.
- **Connect**: (1) one amino acid in the chain changed, so the protein could work differently (in the Siamese cat it stops working at warm temperatures). (2) Accept: candy, not molecules; far bigger than real DNA; real genes are thousands of bases long; real strands twist into a helix.
- **Scoring:** out of 12 (Part A 1, Part B 2, Part C 4, Part D 3, Connect 2), divided by 3 for the 4-point score.